Longevity Drug Distribution's Effects on Population Growth

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1 Introduction

This project aims to describe how the longevity and the application of a longevity drug, with different distribution and intake scenarios, affect the population of different countries. This study is a part of a bigger research that forecasts how different types of longevity interventions might change population trends, work forces and the environment.

2 Abstract

We develop a model that is based on longevity drug with known outcomes on humans that predict population changes by decades. The drug is given to varied groups in the population based on age, economics, education and distribution of other medicine.

This model is tested on several countries, but offers a generic way to predict population changes in a country according to different scenarios.

This project is also accompanied by a theoretical section which offers possible mathematical models for simulating medicine usage and adherence.

3 Methods

3.1 Scenarios

This study covers a diverse set of scenarios.

- **Base Scenarios** - No one takes the medicine, everyone take the medicine.
- **Age-Based Distribution** - The longevity drug is given to different age groups. The medication intake is determined by the percentage of the population in that age range who consistently take medical drugs.
- **Education-Based Distribution** - The longevity drug is given to different education attainment groups, based on each group's medicine intake percentage.
- **Vaccine-Based Distribution** - The longevity drug is given to people in similar distribution to that of vaccinated people in each country.
- **Economic-Based Distribution** - attempts to predict how many people will take the drug if it is free and if it is not (uses vaccine-based distribution as base)

Our models all make the assumption that once a person starts taking the drug, he continues to to so until death.

For each scenario, survival curves are calculated based on the **Saturation Removal Model** under two possible scenarios:

- People who were older than the "minimal age" (according to our distribution) for taking the medicine when it was released into markest.
- people that were younger than the "minimal age" (or weren't born) when it was released

We define t_0 as a person's age when the medication was released (with $t_0 = 0$ for people born after the medication's release). The survival curve for each t_0 is different.

Our Longevity Drug

We simulate one longevity drugs, whose function is decreasing η by 25% from $3.6 \cdot 10^{-6}$ to $2.775 \cdot 10^{-6}$, but the methods used in this research could be used to simulate any change in the saturation removal model, such as increasing X_c.

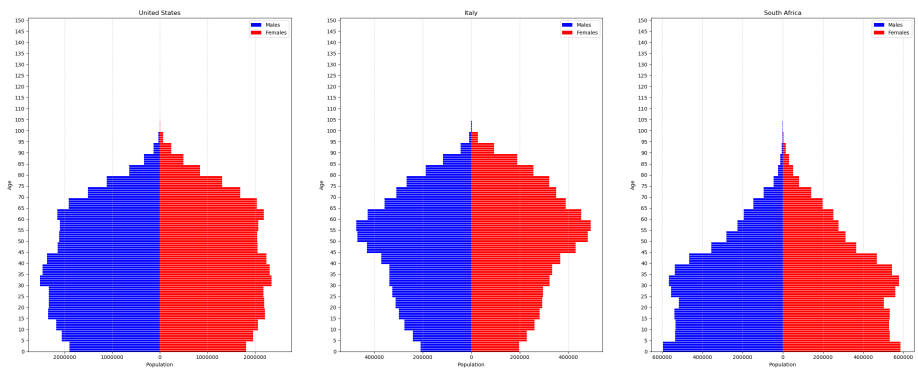
3.2 Population Pyramids

The main method which we use in order to quantify and describe the population of a given country is population pyramids, which showcase the population's distribution by gender and age.

Population pyramids can be divided into three main categories:

- **Stationary** - signifies a balanced population, relatively equal distribution over all different ages
- **Constrictive** - signifies an older population
- **Expansive** - signifies a younger population

For our simulation, we chose countries with different types of population distributions: **United States** (Stationary), **Italy** (Constrictive) and **South Africa** (Expansive) and **Uganda**.



(a) United States (b) Italy (c) South Africa

Figure 1: Population Pyramids

3.3 Leslie Matrix

The Leslie matrix is a population growth model for female population, allowing us to progress the population by one time unit (year in our case) for each time we multiply the matrix.

$$\begin{pmatrix} n_0(t+1) \\ \vdots \\ \vdots \\ \vdots \\ n_K(t+1) \end{pmatrix} = \begin{pmatrix} F_0 & F_1 & \dots & F_K \\ S_0 & 0 & \dots & 0 \\ 0 & S_1 & \dots & 0 \\ \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & \dots & S_K \end{pmatrix} \cdot \begin{pmatrix} n_0(t) \\ \vdots \\ \vdots \\ \vdots \\ n_K(t) \end{pmatrix}$$

Where:

- K - maximum age
- $n_i(t)$ - amount of females at age i at year t

- F_i - fertility rate for female of age i (average amount of births per year)
- S_i - survival rate from year i to year i+1

In order to create a population pyramid that represents all of the population and not only the female part of it, we created the following model:

$$\begin{pmatrix} n_{0,F}(t+1) \\ \vdots \\ \vdots \\ \vdots \\ n_{K,F}(t+1) \\ \hline n_{0,M}(t+1) \\ \vdots \\ \vdots \\ \vdots \\ n_{K,M}(t+1) \end{pmatrix} = \left(\begin{array}{cccc|cccc} F_{0,F} & F_{1,F} & \dots & F_{K,F} & 0 & 0 & \dots & 0 \\ S_{0,F} & 0 & \dots & 0 & 0 & 0 & \dots & 0 \\ 0 & S_{1,F} & \dots & 0 & 0 & 0 & \dots & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & \dots & S_{K,F} & 0 & 0 & \dots & 0 \\ \hline F_{0,M} & F_{1,M} & \dots & F_{K,M} & 0 & 0 & \dots & 0 \\ 0 & 0 & \dots & 0 & S_{0,M} & 0 & \dots & 0 \\ 0 & 0 & \dots & 0 & 0 & S_{1,M} & \dots & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & \dots & 0 & 0 & 0 & \dots & S_{K,M} \end{array}\right) \cdot \begin{pmatrix} n_{0,F}(t) \\ \vdots \\ \vdots \\ \vdots \\ n_{K,F}(t) \\ \hline n_{0,M}(t) \\ \vdots \\ \vdots \\ \vdots \\ n_{K,M}(t) \end{pmatrix}$$

Where:

- $n_{i,G}(t)$ - amount of people of gender G at age i at year t
- $S_{i,G}$ - survival rate from age i to age i+1 for gender G
- $F_{i,G}$ - fertility rate of gender G birth per female

4 Our Simulation

4.1 Distribution Models

In the majority of our models each person either starts the medication at age t_0, or never does.

4.1.1 Age-Based Distribution Models

Our age-based model are different, being based on the knowledge of the fraction of people who take the drug, q, at a certain age range $[x_0, x_1]$ of length x.

We offer three different models, but only use one, **the absolute model**, in our actual simulation for sake of brevity.

Technicalities:

- Given an age range that has fraction q. If t ($t \leq q$) of the people had already taken the drug before said age range, the actual percentage that we will use for our calculations for the age range is: $q = q' \cdot (1 - t) + t \rightarrow q' = \frac{q-t}{1-t}$
- If an age range is not bounded, $[x_0, \infty)$, for sake of our calculations we use $x = 85 - x_0 + 1$

Absolute Model

Given a fraction q and an age range $[x_0, x_1]$, each person has probability q to start taking at exactly age x_0.

Interaction with t_0:

- If $t_0 > x_1$: the person can't take the drug at range $[x_0, x_1]$
- If $x_0 \leq t_0 \leq x_1$: the person will have the same probability to start at age t_0 instead of age x_0

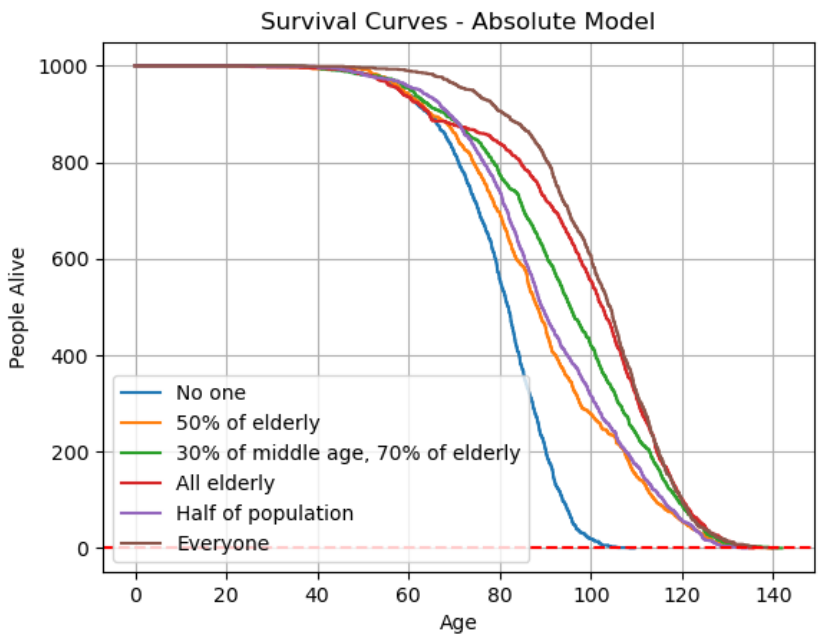


Figure 2: Survival Curve of the Absolute Model Under Different Scenarios

Probabilistic Model with Equal Probabilities

This model attempts to choose an equal probability p for each year in the age range.

A given person has probability p to start taking the drug at age x_0. If he didn't start at x_0, he starts it at age $x_0 + 1$ with probability p, and so on until age x_1.

We choose $p = 1 - (1 - q)^{1/x}$, as the probability of a person starting to take the drug within the age range with such p is indeed q (P(drug isn't taken for the entire age range) $= 1 - q = (1 - p)^x$).

Interaction with t_0: probability is 0 as long as age $< t_0$

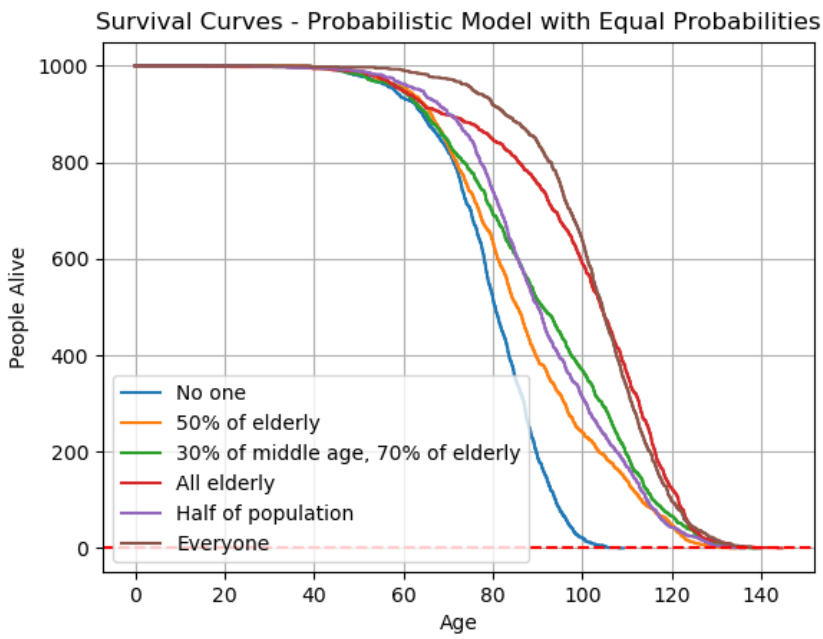


Figure 3: Survival Curve of the Equal Probabilities Model Under Different Scenarios

Probabilistic Model with Uniform Distribution

This model attempts to choose a different probability for each year in the age range, such that a given person will have equal probabilities to start at any age in the age range $[x_0, x_1]$.

We do so by choosing probability $p_{x_0+i} = \frac{q}{x-q\cdot i}$ for age $x_0 + i \leq x_1$, resulting in a uniform distribution over the ages.

The probability is chosen so $\frac{q}{x}$ of the people to start taking the drug in each year of the age range

Before the year $x_0 + i$ an average of $\frac{iq}{x}$ have already started taking the drug, resulting in $p_{x_0+i} = \frac{\frac{q}{x}}{1-\frac{q}{x}\cdot i} = \frac{q}{x-q\cdot i}$

Interaction with t_0: $p_i(t_0) = \begin{cases} 0 & i < t_0 \\ p_i & i \geq t_0 \end{cases}$

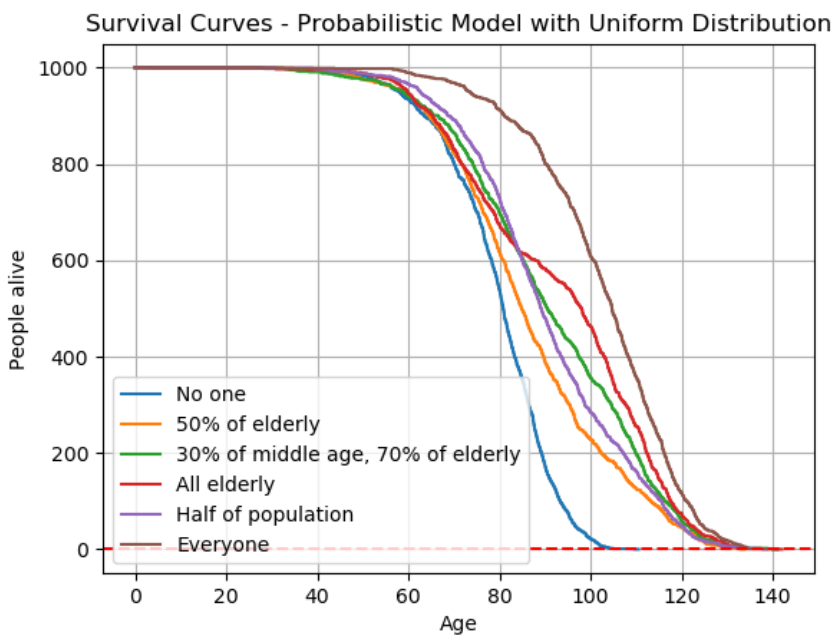


Figure 4: Survival Curve of the Uniform Distribution Model Under Different Scenarios

4.2 Population Pyramids

The data that we use[1] shows the population distribution over 5 years intervals (ages 0-4, ages 5-9, and so on).

Since our calculations required an estimate for each exact age, we divided the population count in each age range uniformly within it.

We neglect the fact that the interval 95+ in the original data is bigger than 5 years, only distributing the amount over the age range 95-99.

4.3 Leslie Matrices

In order to calculate a leslie matrix we need both fertility rates and survival rates, which we calculate as follows:

Fertility rates - we used known fertility rates per country (for ages 15-49), and distributed them between male and female births using the countries' male-to-female birth ratios[3].

We use constant fertility rates over the progression of the years. In reality the fertility ranges would change, but we predicting how they will change and be affected by the drug is beyond the scope of this project.

Survival rates - the product of two components:

- Aging-related survival rate ("Intrinsic") - rate of surviving for i to i+1 according to the survival curve. This rate is changed by drug intake and distribution.
- Non-Aging related survival rate ("Extrinsic") - rate of non-aging related death, different for each country.

"Intrinsic" survival rates

As stated previously, they are calculated by observing how many people are expected to survive using our survival curves.

Assume that the drug was released at year y_0, we determine the survival rate S_i for year $y_j = y_0 + j$ as follows:

A person aged i would have been of age $i - j$ at the drug's release, so $t_0 = \max\{0, i - j\}$.

Since the survival curve changes based on t_0's value, the "intrinsic" survival rate for age i and year y_j needs to be calculated using a survival curve composed of people with $t_0 = \max\{0, i - j\}$. The value itself would be the amount of people alive at age i+1 divided by the amount of people alive at age i.

The main implication is that for the survival rates slightly change every year for people of age $\geq j$ (this obviously stops being relevant when all of the initial population dies)

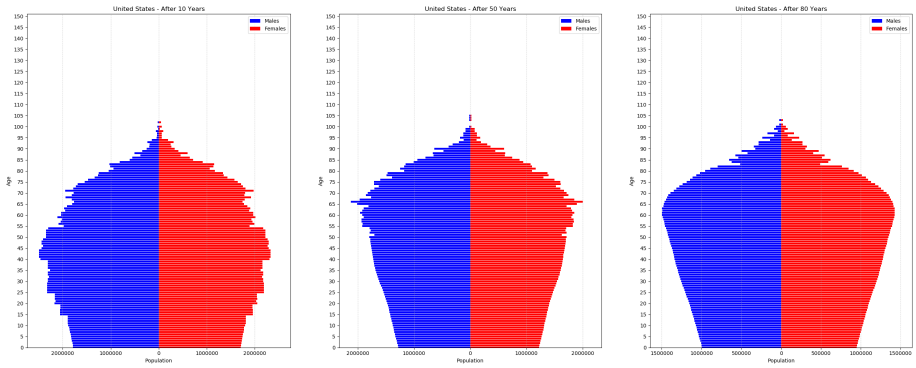
We approximate the "extrinsic" **mortality rates** as follows:

- Divide the population into age ranges: 0, 1-15, 15+
- For each one of those calculate the expected intrinsic mortality rate according to the survival curve for a specific age: 0, 10, 25 (for lack of better data). We purposely choose ages for which intrinsic mortality is very rare.
- subtract the expected intrinsic mortality rate from the countries' actual mortality rate[4]

5 Results

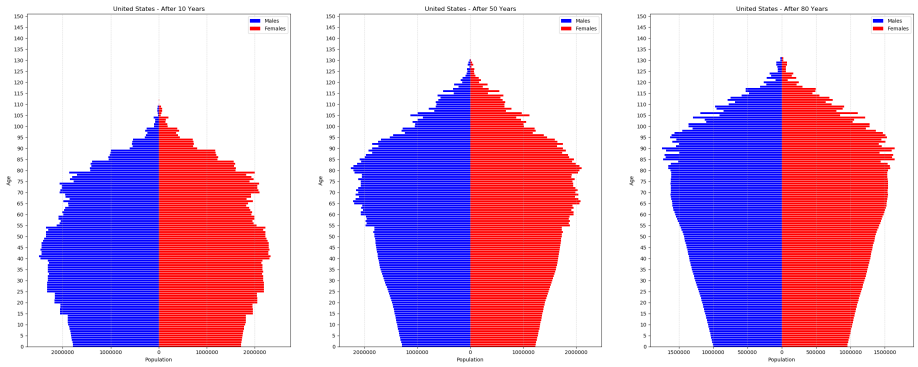
5.1 Base Scenarios

5.1.1 United States



(a) After 10 Years (b) After 50 Years (c) After 80 Years

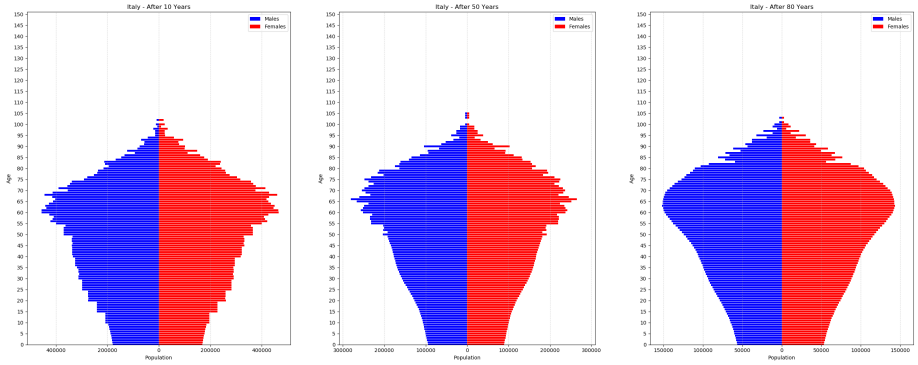
Figure 5: United States Population Pyramid Over the Years - No Longevity Drug



(a) After 10 Years (b) After 50 Years (c) After 80 Years

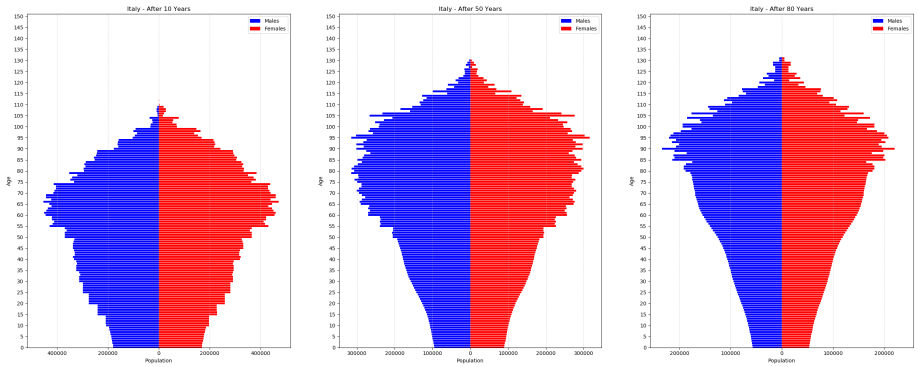
Figure 6: United States Population Pyramid Over the Years - With Longevity Drug

5.1.2 Italy



(a) After 10 Years (b) After 50 Years (c) After 80 Years

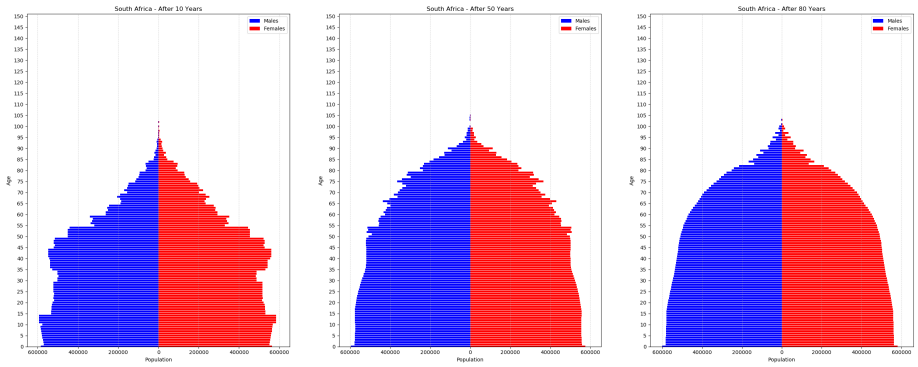
Figure 7: Italy Population Pyramid Over the Years - No Longevity Drug



(a) After 10 Years (b) After 50 Years (c) After 80 Years

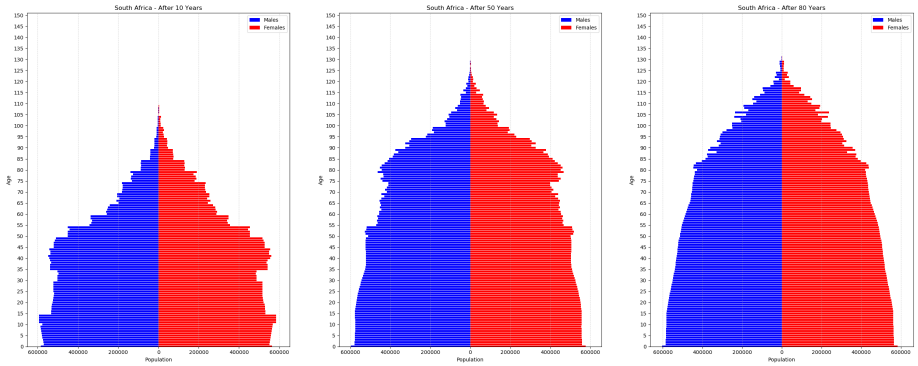
Figure 8: Italy Population Pyramid Over the Years - With Longevity Drug

5.1.3 South Africa



(a) After 10 Years (b) After 50 Years (c) After 80 Years

Figure 9: South Africa Population Pyramid Over the Years - No Longevity Drug



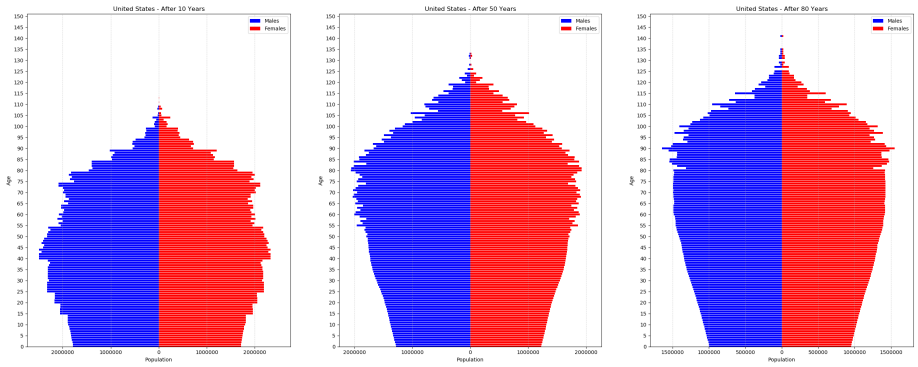
(a) After 10 Years (b) After 50 Years (c) After 80 Years

Figure 10: South Africa Population Pyramid Over the Years - With Longevity Drug

5.2 Age-Based Distribution

5.2.1 United States

Only Elderly



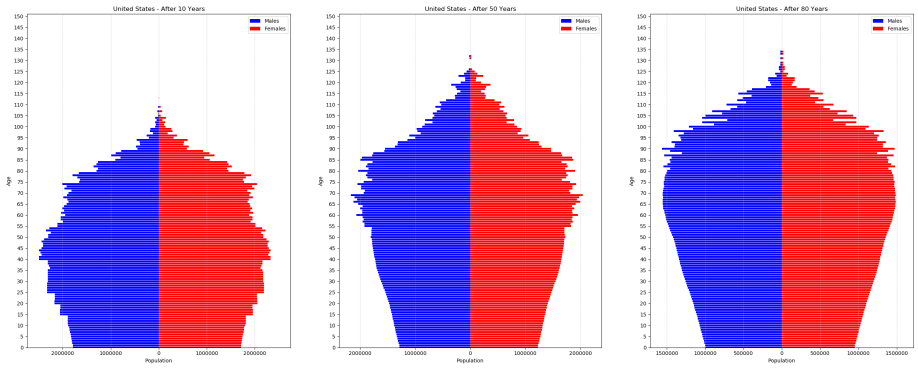
(a) After 10 Years (b) After 50 Years (c) After 80 Years

Figure 11: United States Population Pyramid Over the Years - Only Elderly (65+)

Prescription Drug Use by Age Range[6]

Age Range	Medical Intake Percentage
20-39	35%
40-64	65%
65+	95%

We use the percentages on said age ranges to simulate a distribution of our drug.

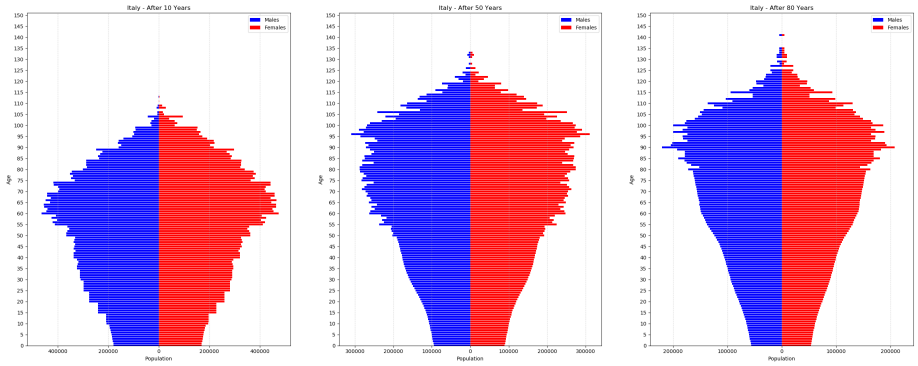


(a) After 10 Years (b) After 50 Years (c) After 80 Years

Figure 12: United States Population Pyramid Over the Years - By Prescription Drug Intake

5.2.2 Italy

Only Elderly



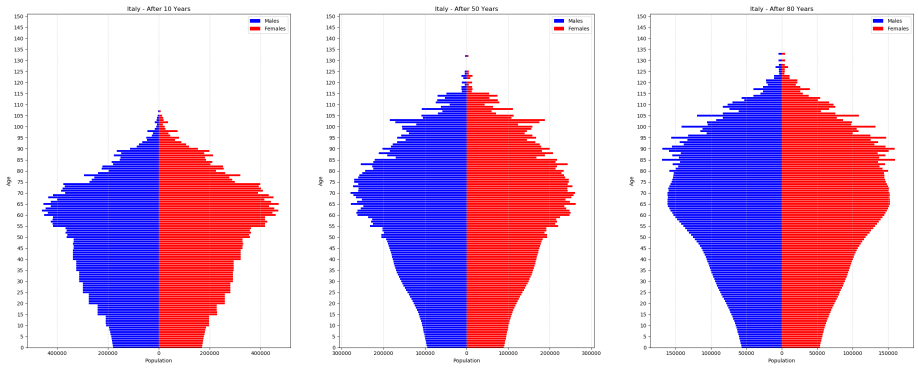
(a) After 10 Years (b) After 50 Years (c) After 80 Years

Figure 13: Italy Population Pyramid Over the Years - Only Elderly (65+)

Self-Reported Use of Prescription Drugs by Age Range[5]

Age Range	Medical Intake Percentage
20-39	35%
40-64	65%
65+	95%

We use the percentages on said age ranges to simulate a distribution of our drug.

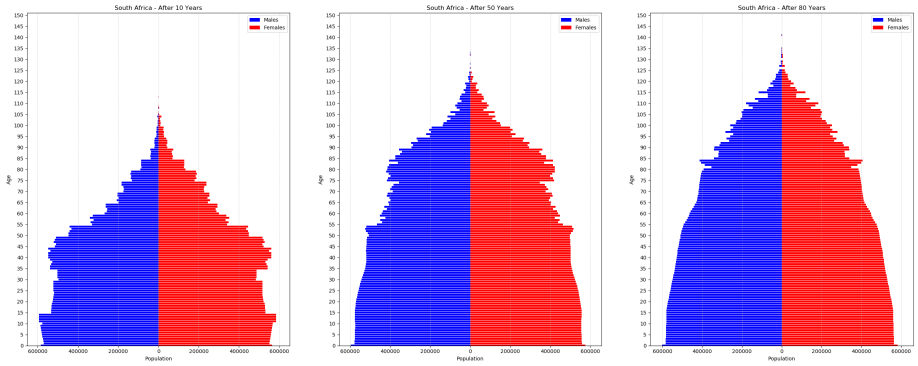


(a) After 10 Years (b) After 50 Years (c) After 80 Years

Figure 14: Italy Population Pyramid Over the Years - By Prescription Drug Intake

5.2.3 South Africa

Only Elderly



(a) After 10 Years (b) After 50 Years (c) After 80 Years

Figure 15: South Africa Population Pyramid Over the Years - Only Elderly (65+)

5.3 Education-Based Distribution

We simulate drug distribution based on self-reported use of medicine by medical attainment (for ages ≥ 18).

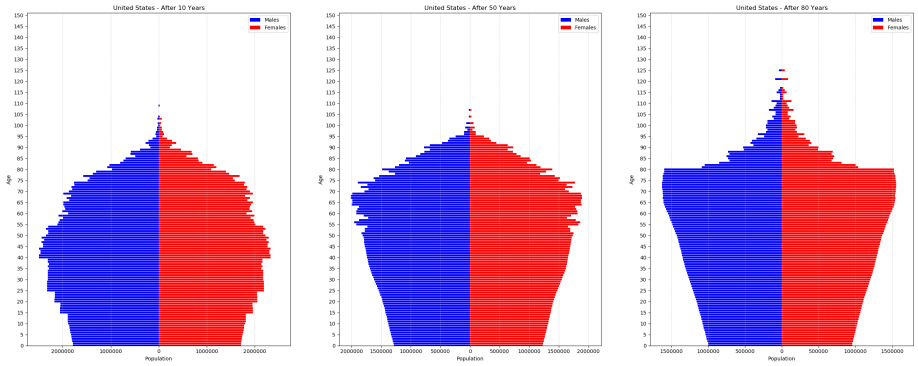
For this simulation we use an absolute model, where people either start taking at age t_0 or never at all.

USA

Use of Prescription Drugs by Educational Attainment[6][7]

Education Attainment	Medical Intake Percentage
No high school	57%
High school	58%
Some college	57%
College	61

We use the percentages on said educational levels (according to each groups' percentage in the solution) to simulate a distribution of our drug.



(a) After 10 Years (b) After 50 Years (c) After 80 Years

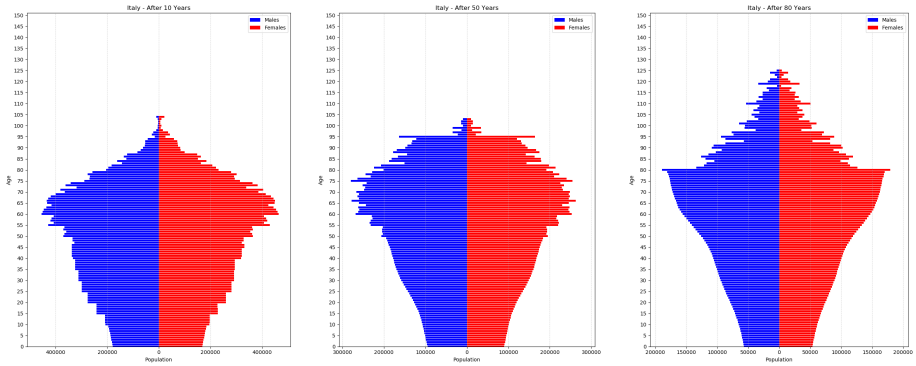
Figure 16: United States Population Pyramid Over the Years - By Prescription Drug Intake

5.3.1 Italy

Self-Reported Use of Non-Prescription Drugs by Educational Attainment[9][8]

Education Attainment	Medical Intake Percentage
At Most Lower Secondary	16.1%
Upper-Secondary and Post-Secondary	20.2%
Tertiary Secondary	25.0%
College	61

We use the percentages on said educational levels (according to each groups' percentage in the solution) to simulate a distribution of our drug.



(a) After 10 Years (b) After 50 Years (c) After 80 Years

Figure 17: Italy Population Pyramid Over the Years - By Non-Prescription Drug Intake

5.4 Vaccination-Based Distribution

We shall observe the percentage of people fully-vaccinated for COVID-19 in each country[10], and distribute our longevity drug to similar percentages.

Once again, this simulation uses an absolute model where people either start taking the drug at age t_0 or never at all.

We showcase the difference in survival curve between the different countries under said scenario:

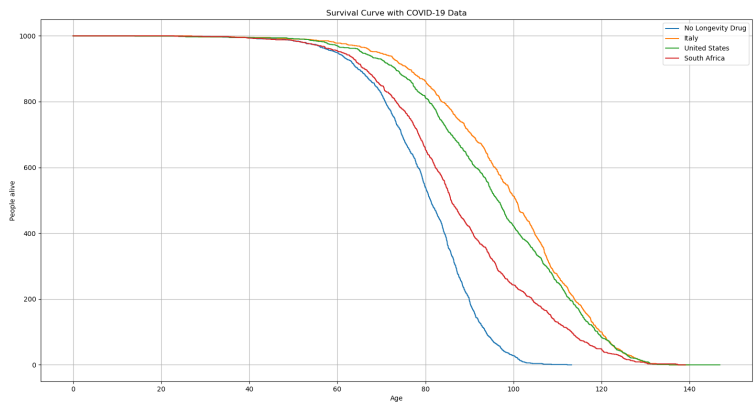


Figure 18: Survival Curves - By COVID-19 Distribution

5.5 Economics' Effect on Distribution

According to surveys, 3 out of 10 Americans report not taking medicine as prescribed because of its price[11].

When shall use the COVID-19 vaccinations usage percentages in the United States, 67.53%, for reference.

We chose the COVID-19 vaccines specifically because they were distributed for free, and by taking $\frac{7}{10} \cdot 67.53\%$ we can simulate medical adherence to a "pricier" drug compared to a free one.

We showcase the effect of the drug's price on the survival curve and on the overall population size in the United States

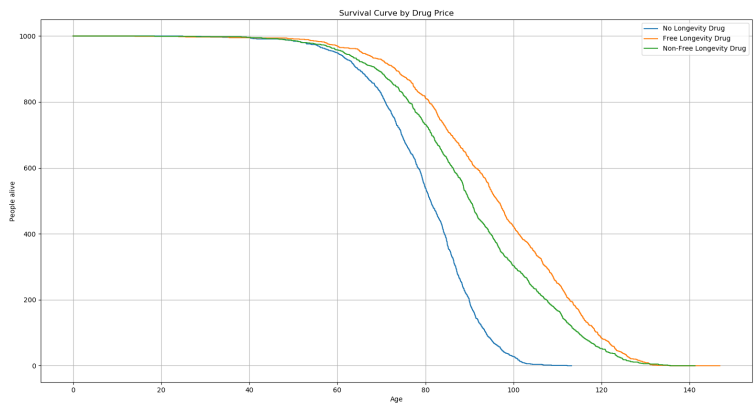


Figure 19: United States Survival Curves by Drug Price

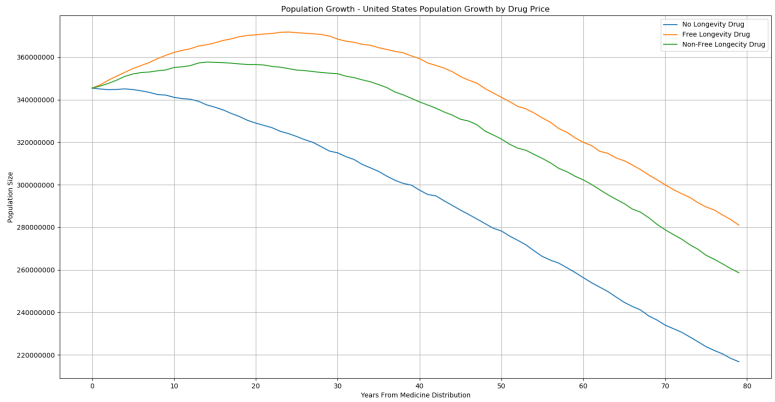


Figure 20: Population Growth of United States by Drug Price

5.6 Population Growth Over the Different Scenarios

United States

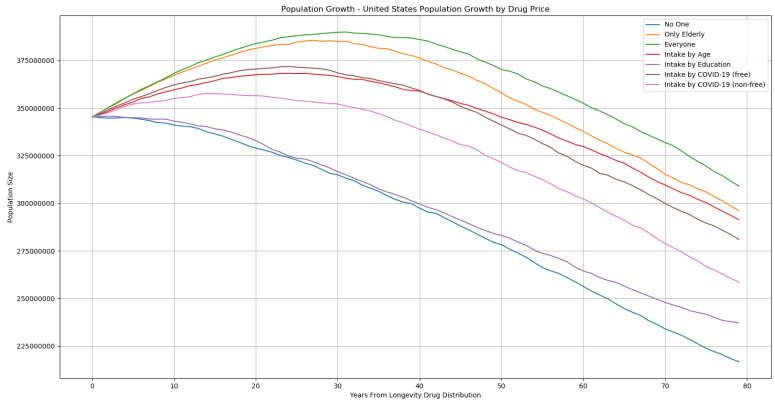


Figure 21: United States Population Growth

Italy

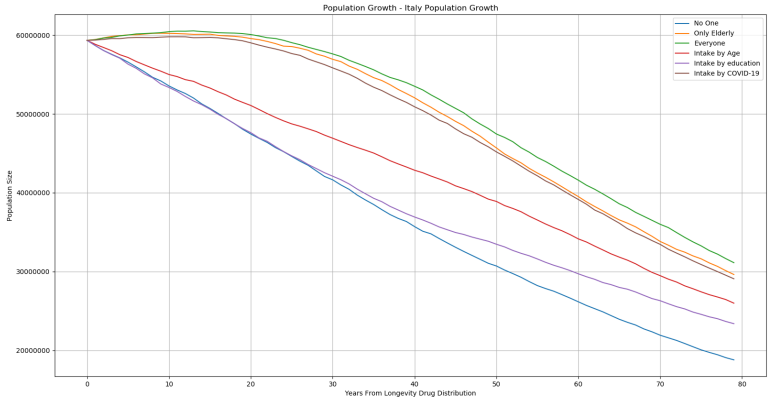


Figure 22: Italy Population Growth

South Africa

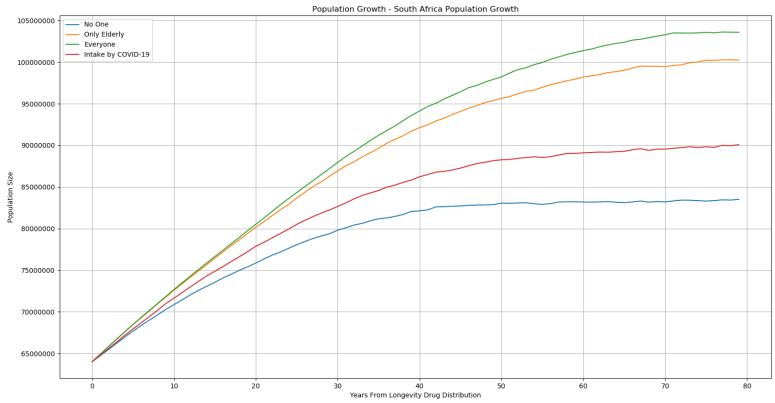


Figure 23: South Africa Population Growth

6 Consistency of Medicine Intake

We offer some rough mathematical models to generalize scenarios where continued intake of the medicine isn't guaranteed.

6.1 Naive Model

We offer where each person has probability p to start taking the medication each year. Once he has started it, he has probability q every year to stop it.

Define I_t to be the indicator which is 1 if the person is taking the drug in year t.

We observe a medication which changes the base eta value η_0 (when $I_t = 0$) into η' (when $I_t = 1$), such that $\eta(t) = (\eta' - \eta_0)I_t + \eta_0$.

We shall prove that the probability to take the drug at age t is $P(I_t = 1) = p \cdot \frac{1-(1-q-p)^{t+1}}{q+p}$ for $t \geq 1$ using induction.

Base:

For t=1, $P(I_1 = 1) = P(I_1 = 1 \mid I_0 = 1) \cdot P(I_0 = 1) + P(I_1 = 1|I_0 = 0) \cdot P(I_0 = 0) = (1-q) \cdot p + p \cdot (1-p) = (1-q-p) \cdot p + p = p \cdot \frac{1-(1-q-p)^2}{q+p}$,

Step: We know that $P(I_t = 1) = p \cdot \frac{1-(1-q-p)^{t+1}}{q+p}$, and shall prove that $P(I_{t+1} = 1) = p \cdot \frac{1-(1-q-p)^{t+2}}{q+p}$

$$\begin{aligned}
P(I_{t+1} = 1) &= P(I_{t+1} = 1|I_t = 0) \cdot P(I_t = 0) + P(I_{t+1} = 1|I_t = 1) \cdot P(I_t = 1) \\
&p \cdot (1 - P(I_t = 1)) + (1-q) \cdot P(I_t = 1) = (1-q-p) \cdot P(I_t = 1) + p \\
&= (1-q-p) \cdot p \cdot \frac{1-(1-q-p)^t}{q+p} + p = p \cdot \frac{1-q-p-(1-q-p)^{t+1}+q+p}{q+p} \\
&= p \cdot \frac{1-(1-q-p)^{t+1}}{q+p}
\end{aligned}$$

6.2 Optional Improvements

Existing problems in the naive model:

- p and q don't change based on the person's drug intake history
- η only has two possible values

We change q to be the probability of a person not taking the drug in a specific time unit.

We define for each person a **consistency factor** that changes over time, c. It is a constant that is dependent on the time intervals in which the drug was taken (a possible option is a sort of weighted average based on each interval's length and recency).

We define p and q as functions dependent on c. For example: the more a person forgets to take his medicine (represented by q), c will become smaller, and q will become bigger, and vice versa.

p(c) and q(c) also change based on the following factors:

- Country
- Age
- Gender
- Education attainment
- Economical status

Instead of only having two possible values, we define $\eta(t, c)$ as a function dependent on the age and consistency factor of the person, with minimum value η_0 and maximum value η_{max}. $\eta(t, c)$ should be closer to η_{max} the more consistent a person is.

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